

STEPHEN GAVIN SEARS

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Software Engineer — GPU Programming, Graphics & Machine Learning

Master's student in Computer Graphics and Game Technology at the University of Pennsylvania, with a B.S. in Computer Science from Haverford College. Experience spans GPU programming (a CUDA path tracer with BVH acceleration and denoising), applied machine learning (LLM tool-calling agents, custom ML pipelines), and real-time graphics and tools development in C++, Unity, and Python. Git and GitHub enthusiast.

EDUCATION

MSE, Computer Graphics & Game Technology — University of Pennsylvania, Philadelphia, PA *Expected May 2027*

Relevant Coursework: GPU Programming, Applied Machine Learning, Interactive Graphics, Procedural Computer Graphics, Advanced Topics in Computer Graphics, Modern Convex Optimization

BS, Computer Science (Visual Studies Minor) — Haverford College, Haverford, PA *Aug. 2021 – May 2025*

Relevant Coursework: Machine Learning, Computer Graphics, Analysis of Algorithms, Data Structures, Computer Organization, Data Science, Discrete Mathematics, Computer Animation, Game Design

PROJECTS

GPU Path Tracer — C++, CUDA, OpenGL — [Repo Link](#) *Penn CIS 5650, Fall 2026*

- Built a physically based GPU path tracer in C++/CUDA with multiple importance sampling and next event estimation (MIS + NEE), GGX microfacet and refractive materials, HDRI lighting, thin-lens depth of field, and texture/bump mapping for glTF
- Implemented a median-split BVH that cut render time on a 70k triangle mesh by 960x, and built heatmap and leaf-node debug views to profile traversal cost
- Added stream compaction (thrust) and material sorting (CUB radix sort), reaching up to 1.9x speedup in open scenes
- Integrated Intel Open Image Denoise on the GPU with albedo and normal buffers for interactive denoised previews

Gaussian Fluids — Houdini Plugin (C++/C) — [Repo Link](#) *Penn, Advanced Topics in Computer Graphics*

- Implemented the Gaussian Fluids research paper as a Houdini plugin, representing a velocity field as Gaussians to simulate fluid motion that preserves more vorticity and uses less memory than typical grid-based solvers

Virtual Aquarium — Procedural WebGL2 Renderer — [virtual-aquarium-5660.vercel.app](#) *Penn CIS 5660, Fall 2025*

- Designed a procedural fish generator that extrudes parameterized ring loops along a body spine, with 20+ real-time parameters and per-frame pivot-based transforms

WORK EXPERIENCE

Software Intern — *Astronics Test Systems, Orlando, FL* *June 2026 – Aug. 2026*

- Built an end-to-end voice interface for RF spectrum analyzers: faster-whisper transcription (Python/FastAPI), a swappable local (Gemma 4 E4B) or hosted LLM, and a Model Context Protocol (MCP) server issuing tool calls into instrument control
- Evaluated agent behavior on engineer-collected task workflows, including multi-step planning and refusal of out-of-scope requests
- Compared local and hosted models on latency and hardware load
- Built a WPF .NET 8 C# diagnostic tool with an in-memory SQLite database consolidating hardware, firmware, and system logs; used agentic AI tools under ITAR/CUI data-handling requirements

Teaching Assistant, Machine Learning — *Haverford College, Haverford, PA* *Jan. 2025 – May 2025*

- Held weekly TA hours and graded assignments on theoretical AI/ML concepts, Python, TensorFlow, and scikit-learn

The HyperReal Theater — *Prof. Aline Normoyle's Lab — Bryn Mawr College, Bryn Mawr, PA* *Aug. 2024 – Dec. 2025*

- Developed IK solvers in Unity C#, rigged and textured 3D models, and calibrated motion-capture equipment for professional demos

Data Analyst & Power BI Specialist — *Sears Mfg. Co., Davenport, IA* *May 2025 – Aug. 2025*

- Built 6 automated Power BI reports and Data Lake/Warehouse pipelines (Transact-SQL, PySpark) for supply-chain and cost-engineering visibility

TECHNICAL SKILLS

Languages: C++, C, CUDA, Python, C#, GLSL, JavaScript, TypeScript, Transact-SQL

Graphics & GPU: OpenGL, WebGL2, CUDA, Unity, path tracing, BVH acceleration, stream compaction, shader authoring

ML & LLM: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, faster-whisper, Model Context Protocol (MCP), tool calling

Backend & Data: REST APIs (FastAPI, ASP.NET Core), PySpark, SQLite, Docker, Linux (Ubuntu, WSL)

Tools: Git/GitHub, Visual Studio, VS Code, RenderDoc, Houdini, Jira, Confluence